# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years of experience building **systems and engineering tools**, **C++/Python applications**, data pipelines, automated tests, and **reliable cross-platform software** for production workloads.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

## Experience

**USC Dornsife** | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and error handling**, cloud storage, CI/CD releases, and production monitoring supporting dependable internal software workflows and technical operations.
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant microservices** with explicit state transitions and failure isolation, preserving recoverable background processing reliably under sustained production workloads.

**Grad8** | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured automated testing**, model switching, fallback paths, persistent results, and reviewer controls supporting reliable production workflows and troubleshooting practices.
- Processed 1K+ files per batch using **multithreaded validation and fault isolation**, preserving responsive workflows while troubleshooting heterogeneous PDF, DOCX, and ZIP inputs under concurrent model execution reliably.

**USC AD Labs** | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **reproducible evaluation pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and controlled experiments across diverse scenarios reliably.
- Improved closed-loop planning stability by 15% through a **simulation-based validation model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex scenarios reliably.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer - (SecureSync)* | *India*
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.
- Improved API response times by 30% through **object-oriented service components**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably.

**Cyber Security and Data Mining: Detecting Malicious URL** **May 2023 – August 2023**
*Data Analyst - Research Fellowship* | *India*
- Achieved 96%+ classification accuracy across 650K+ URLs through a **reproducible test pipeline**, comparing Random Forest, CNN, RNN, LSTM, Bi-LSTM, and GRU models for malware and phishing detection reliably.
- Engineered 21 behavioral indicators covering URL length, IP presence, and suspicious keywords, translating noisy inputs into **validated classification features** that exposed phishing and malware patterns reliably at scale.

## Skills

**Languages:** C++, Python, Java, SQL, JavaScript, TypeScript, Bash
**Systems & Fundamentals:** Operating Systems, Object-Oriented Programming, Data Structures, Algorithms, Multithreading, Concurrent Programming, Simulation, System Design, JSON, XML
**Data & Applications:** Data Pipelines, ETL, PostgreSQL, MySQL, MongoDB, AWS RDS, Data Modeling, Query Optimization, React, Data Visualization
**Testing & Quality:** Unit Testing, Integration Testing, Automated Testing, Regression Testing, PyTest, Mockito, Debugging, Root-Cause Analysis, Fault Isolation, Performance Testing
**Tools & Delivery:** Linux, Windows, Git, GitHub, Jenkins, Docker, Maven, Node.js, CI/CD, Version Control, Build Automation
**Engineering Practices:** Software Development Lifecycle, Requirements Analysis, Technical Documentation, Cross-Platform Development, Code Review, Error Handling, Production Reliability

## Projects

**Storage Systems Diagnostic Toolkit - Design Concept**
- Designed a **storage-systems diagnostic toolkit** concept in C++ and Python, generating configurable I/O workloads, collecting latency and throughput telemetry, detecting regressions, visualizing results, and validating behavior across operating-system scenarios.

**CPU-Scheduling Simulator** | Live
- Built a **CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios reliably.

**Chat2SQL** | Live
- Extended a **cross-platform Java application** supporting 10+ database systems, using Docker, Maven, Node.js, and Yarn for structured validation, schema exploration, query workflows, local debugging, and repeatable deployment.